# Ioannis Panitsas

Department of Electrical and Computer Engineering, Yale University
17 Hillhouse Avenue, New Haven, CT 06511, USA
ioannis.panitsas@yale.edu | panitsasi.github.io

## Research Interests

My research lies at the intersection of next-generation cellular networks and applied machine learning. I develop learning-based methods for the management, optimization, and security of next-generation wireless systems, combining algorithm design with systems evaluation on production-grade testbeds.

## Education

**Yale University**, New Haven, CT, USA — 08/2022 – 08/2027 *(expected)*
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Leandros Tassiulas

**Yale University**, New Haven, CT, USA — 08/2022 – 05/2025
M.Phil. in Electrical and Computer Engineering

**Yale University**, New Haven, CT, USA — 08/2022 – 05/2024
M.Sc. in Electrical and Computer Engineering

**Hellenic Army School of Military Engineering**, Loutraki, Greece — 03/2021 – 05/2022
2nd Lieutenant, Engineering Corps

**University of Patras**, Patras, Greece — 09/2016 – 07/2021
Integrated B.Sc. and M.Sc. in Electrical and Computer Engineering
Advisor: Prof. Dimitrios Serpanos

## Publications

### Conference Publications

[C1] **I. Panitsas**, T. Atalay, D. Stojadinovic, A. Stavrou, and L. Tassiulas, "SlicePilot: Demystifying Network Slice Placement in Heterogeneous Cloud Infrastructures," *IEEE International Conference on Computer Communications (INFOCOM)*, 2026.

[C2] **I. Panitsas**, I. Ofeidis, and L. Tassiulas, "FedJam: Multimodal Federated Learning Framework for Jamming Detection," *IEEE International Conference on Computer Communications (INFOCOM)*, 2026.

[C3] A. Feng, A. Varvarigos, **I. Panitsas**, D. Fernandez, J. Wei, Y. Guo, J. Chen, A. Maatouk, L. Tassiulas, and R. Ying, "TelecomTS: A Multi-Modal Observability Dataset for Time Series and Language Analysis," *International Conference on Machine Learning (ICML)*, 2026.

[C4] **I. Panitsas**, T. Atalay, D. Stojadinovic, A. Stavrou, and L. Tassiulas, "5GC-Bench: A Framework for Stress-Testing and Benchmarking 5G Core VNFs," *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.

[C5] **I. Panitsas**, A. Mudvari, A. Maatouk, and L. Tassiulas, "A Deep and Transfer Learning Approach for Handover Management in O-RAN," *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.

[C6] **I. Panitsas**, Y. Yigit, L. Tassiulas, L. Maglaras, and B. Canberk, "JamShield: A Machine Learning Detection System for Over-the-Air Jamming Attacks," *IEEE International Conference on Communications (ICC)*, 2025.

[C7] **I. Panitsas**, A. Mudvari, and L. Tassiulas, "D2Q Synchronizer: Distributed SDN Synchronization for Time-Sensitive Applications," *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, 2025.

[C8] Y. Yigit, **I. Panitsas**, L. Maglaras, L. Tassiulas, and B. Canberk, "Cyber-Twin: Digital Twin-Boosted Autonomous Attack Detection for Vehicular Ad-Hoc Networks," *IEEE International Conference on Communications (ICC)*, 2024.

**Manuscripts Under Review**

[S1] **I. Panitsas** Ajay Rajkumar, and L. Tassiulas, "HARMONY: Towards Hierarchical Conflict Management in Open RAN," under review.

[S2] **I. Panitsas** and L. Tassiulas, "TimeRAN: Multi-Task Time-Series Learning for Next-Generation Radio Access Networks," under review.

[S3] **I. Panitsas**, P. Gajjar, E. Ojo, V. K. Shah, and L. Tassiulas, "PRISMA: Multivariate Anomaly Detection and Root-Cause Diagnosis for Open RAN," under review.

## Research Experience

**Graduate Research Assistant** 08/2022 – Present
Yale University, New Haven, CT, USA

- Develop machine learning methods for network management, optimization, control and security.

**Research Intern**, RAN Technology Research Group 06/2026 – 08/2026
AT&T Labs, Bedminster, NJ, USA

- Designed hierarchical conflict management algorithms for Open RAN deployments.

**Wireless Systems Research Intern**, RAN Platforms Research 01/2026 – 05/2026
Nokia Bell Labs, Murray Hill, NJ, USA

- Developed next-generation AI-RAN architectures and neural receiver implementations for improving uplink 5G performance on GPU-accelerated inline processing pipelines.

**Research Scientist Intern** 07/2025 – 08/2025
A2 Labs, Arlington, VA, USA

- Designed a resource orchestrator for efficient slice placement across cloud infrastructures, and a benchmarking framework for testing and profiling network functions of the 5G Core.

**Research Scientist Intern** 07/2024 – 08/2024
Kryptowire Labs, Arlington, VA, USA

- Developed *SlicePilot*, a multi-agent reinforcement learning-based system for optimal network slice placement and resource allocation across heterogeneous cloud infrastructures.

**Embedded Software Engineer Intern** 08/2020 – 10/2020
Renesas Electronics, Patras, Greece

- Performed link-layer penetration testing and security evaluation of the Bluetooth Low Energy 5.0 protocol, identifying protocol-level vulnerabilities through crafted over-the-air exploits.

## Research Proposals

Contributed to the preparation of multiple research proposals submitted to NSF, the Army Research Laboratory, the Office of Naval Research, the Air Force Research Laboratory, and NTIA, including a DURIP instrumentation proposal and a joint white paper with Nokia Bell Labs.

## Teaching Experience

*Teaching Assistant, Department of Electrical and Computer Engineering, Yale University.*

| | |
|---|---|
| *Introduction to Communications and Control* (ECE 2020) | Fall 2026 |
| *Introduction to Computing for Engineers and Scientists* (ENAS 130) | Spring 2025 |
| *Introduction to Communications and Control* (EENG 202) | Fall 2024 |
| *Introduction to Computing for Engineers and Scientists* (ENAS 130) | Spring 2024 |

Responsibilities across these courses included leading office hours, preparing and grading problem sets and examinations, and advising students on course projects.

## Research Workshops

**CyberPowder Fellows Program**, University of Utah, Salt Lake City, UT 2025
On-site research workshop on the POWDER wireless testbed

**Open AI Cellular Workshop**, Mississippi State University, Starkville, MS 2024
Hands-on experimentation with the OAIC open-source RAN framework

**Young Gladiators Workshop**, Northeastern University, Boston, MA 2024
On-site research workshop on the Colosseum wireless network emulator

## Professional Service

### Journal Reviewer

- IEEE Communications Magazine, IEEE Transactions on Cognitive Communications and Networking, IEEE Transactions on Information Forensics and Security, IEEE Transactions on Machine Learning in Communications and Networking, IEEE Transactions on Vehicular Technology, IEEE Wireless Communications Letters, IEEE/ACM Transactions on Networking.

### Conference Reviewer

- ACM SIGMETRICS, IEEE Dynamic Spectrum Access Networks (DySPAN), IEEE Global Communications Conference (GLOBECOM), IEEE Military Communications Conference (MILCOM), IEEE Vehicular Technology Conference (VTC).

## Technical Skills

- **Programming:** Python, C/C++, OpenMP, Java, JavaScript, Node.js, SQL, MongoDB, HTML, CSS.
- **Systems and Tooling:** Docker, Kubernetes, Helm.
- **Wireless and 5G Platforms:** OpenAirInterface (OAI), srsRAN, Open5GS, NVIDIA Aerial, Sionna-RK, Sionna-RT, POWDER, Colosseum.

## Languages

Greek (native) | English (fluent)

## References

**Prof. Leandros Tassiulas** *(Ph.D. advisor)*
Department of Electrical Engineering, Yale University
leandros.tassiulas@yale.edu
**Prof. Dimitrios Serpanos** *(B.Sc. advisor)*
Department of Electrical and Computer Engineering, University of Patras
serpanos@ece.upatras.gr